Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	Role of Debt Maturity in Inflation Financing the Fiscal Shocks

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Role of Debt Maturity in Inflation Financing the Fiscal Shocks

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Abstract

This paper adds to the growing literature on optimal financing of large fiscal debts incurred during COVID-19. While existing literature considers a trade-off between using tax revenues or inflation to finance debt, we abstract away from including tax dynamics to look closely at how current and future inflation can be used to lower the real debt burden of governments. A key finding of this paper is the crucial role that a higher duration of debt plays in achieving this, while also smoothing out inflation volatility over time. Therefore, a least-costly form of financial repression is issuing debt of longer-term maturity and using persistently higher inflation to reduce the debt burden over time. This finding is robust to different realistic calibrations of the structural parameters such as price stickiness, returns to scale, elasticities of labor supply, demands, and intertemporal substitution, etc.

Keywords: fiscal shocks, pandemic, monetary policy, duration, inflation, bond prices

Word Count: 5825

1 Introduction

COVID-19 crisis saw a record-high drop in output and rise in government spending all across the world. Naturally, a fall in tax revenues and a rise in primary deficits was financed through historically high level of borrowings. Government debt globally increased to an average of 97% of GDP, with the highest increase coming from developed countries like the United States (128% of GDP) and the United Kingdom (100% of GDP). Such high levels of debt cannot be financed only through the traditional fiscal financing methods of raising tax revenues.

Most literature lays emphasis on tax revenues being a primary source of financing government's increase in deficit (Aiyagari et al. (2002), Schmitt-Grohé and Uribe (2004), Siu (2004))and show that inflation is not that important (Faraglia et al. (2013), Schmitt-Grohé and Uribe (2004)). However, in recent years, many authors have shown that inflation (current and expected) financing might actually be a prevalent and optimal policy (for instance, used in financing 7% and 25% U.S. fiscal shocks (Leeper et al., 2017)). In fact, since this was a policy since 1970s (Bianchi and Ilut, 2017), it helped anchor beliefs of inflation at higher levels post-2008 crisis thus preventing deflation despite being at the zero-lower bound (Bianchi and Melosi, 2017). Teles and Tristani (2021) have shown that inflation financing is an optimal response to rising debt levels, if coupled with long-term maturity.

The model considered in this essay combines the standard New-Keynesian DSGE framework with innovative fiscal constraints, that model a bond-financed fiscal shock and enable me to characterise optimal monetary policy response to such a shock. I follow Sheedy (2022) for this constraint. Unlike most existing literature (Lustig et al. (2008)), I do not include fiscal policy, but rather that the monetary authority needs to satisfy the government budget constraint at all times.

Through the IRFs generated, I conclude that a loss-minimizing policy uses a combination of two channels to finance the initial shock and to change the burden of debt thereafter: short-term nominal interest rates that alter the path of bond prices and long-term interest

rates (yield-to-maturity), and current inflation surprise and path of expected inflation. In the way fiscal shock is defined (see section 2.3), a positive shock corresponds to a tax cut or a rise in transfers from the government to the households. To simplify the analysis and make the inflation response and its transmission through the asset prices (here, bonds) more tractable, I follow Sheedy (2022) and assume that taxes return to steady state immediately after the shock (i.e., it is a one-off shock) and thus, there are no tax dynamics or how fiscal policy reacts to changes in fiscal deficits. The analysis in this paper is purely focused on how monetary policy would optimally react to such shocks, when faced with a trade-off between output gap and inflation stabilization, under commitment.

A key finding of the essay is that higher duration of the debt helps in smoothing inflation over time and it is more welfare improving to use current and future inflation channel rather than bond prices. These findings are consistent with (Hilscher et al. (2022), Angeletos (2002)) who have shown the importance of long debt maturity and the role of a persistent inflation in reducing U.S. government's debt burden and robust to many different realistic calibrations.

The structure of the essay is as follows: section 2 lays out the model, section 3 provides the baseline calibration, and section 4 discusses the transmission mechanisms in the optimal policy and how it changes with different calibrations. Section 5 contains derivations to some of the equations discussed in section 2.

2 Model

For the baseline model, I consider an economy with three sectors - a continuum of representative households who purchase a continuum of differentiated goods (normalized from 0 to 1), produced and sold by monopolistic firms, and a government. Government consists of the fiscal authority which will levy lump-sum taxes, make lump-sum transfers, and issue bonds to raise finance for its primary deficit. These bonds need not be one-period debt. In fact, a long-term bond and the dynamics of its price and the long-term interest rate is carries is one of the main departure points of this model from the standard New-Keynesian framework. As we will see in subsequent sections, it will be important for

optimal inflation and output gap stabilization policies post a fiscal shock. Government also consists of a monetary authority which sets the nominal interest rate on short-term bonds. I take most of the components of the model from Sheedy (2022), while incorporating some details from Leeper and Zhou (2021) for expositional clarity. The agents - both firms and households - are forward looking and have rational expectations.

2.1 Households

Since all the households are identical, I consider a representative household's optimisation problem, which chooses a sequence of $\{C_t, N_t, B_t\}_{t=0}^{\infty}$ that maximises its discounted expected utility subject to a per-period budget constraint. Its preferences are assumed to be of the following form:

$$E_t \sum_{i=0}^{\infty} \beta^i \left[\frac{C_{t+i}^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}} - \frac{N_{t+i}^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} \right]$$

where, the utility function is assumed to be additively separable in consumption (C_t) and labour choice (N_t), σ is the elasticity of intertemporal substitution, and ψ is the Frisch elasticity of labour supply. Since the household consumes a continuum of goods, I assume the basket to be represented by a CES aggregator as below, in line with the standard New Keynesian literature.

$$C_t = \left[\int_0^1 c_{jt}^{\frac{\epsilon-1}{\epsilon}} dj \right]^{\frac{\epsilon}{\epsilon-1}} \qquad \epsilon > 1$$

where, c_{jt} represents consumption of good j consumed at time t and ϵ is the demand elasticity of a differentiated good (it is same for all j goods). Households are constrained by the following budget constraint (following Leeper and Zhou (2021)):

$$C_t + \frac{Q_t}{P_t} B_t + \frac{Q_t^k}{P_t} B_t^k = \frac{W_t}{P_t} N_t + \frac{1+\delta Q_t}{P_t} B_{t-1} + \frac{B_{t-1}^k}{P_t} + \frac{D_t}{P_t}$$

where, I assume that

- Households buy two types of bonds - one period bonds that are in net zero supply just like in standard New-Keynesian framework and long-term bonds for which they are entitled to receive (1, δ, δ^2, δ^3, ...) every subsequent period until maturity. The latter type of bonds are issued by the fiscal section of government sector to finance its

primary deficits. One-period bonds carry a short-term nominal interest rate i_t which is set by the Central Bank (monetary authority of the government sector), while fiscal bonds carry a long-term interest rate j_t, also called yield-to-maturity (YTM). So, when $\delta = 0$ we have standard budget constraint as we only have one-period bonds in the economy. $\delta = 1$, on the other hand, corresponds to a perpetuity.

- Prices of the bonds will reflect the returns received on them.

$$Q_t^k = \frac{1}{1+i_t} \tag{1}$$

$$Q_t = \frac{1}{1+j_t} + \frac{\delta}{1+j_t} + \frac{\delta^2}{1+{j_t}^2} + ... = \frac{1}{1+j_t-\delta} \tag{2}$$

Log-linearised as:

$$q_t = -\frac{1-\beta}{1-\beta\delta} j_t \tag{3}$$

- The following no-arbitrage condition relates the two interest rates (derivation follows from the household's first order conditions and can be found in section 5):

$$j_t = \beta\delta E_t j_{t+1} + (1-\beta\delta) i_t \tag{4}$$

Iterating this equation forward, we get the following relation between yields and future path of short-term nominal interest rates:

$$j_t = (1-\beta\delta) E_t \sum_{T=0}^{\infty} (\beta\delta)^T i_{t+T} \tag{5}$$

Combining equations 3 and 5, gives us a relation between bond prices and future path of short-term nominal interest rates:

$$q_t = -(1-\beta) E_t \sum_{T=0}^{\infty} (\beta\delta)^T i_{t+T} \tag{6}$$

- Parameter δ is closely related to a bond's Macaulay duration (DU) (weighted average time by which the bondholder will receive all associated cash flows (CF_t)). It is

defined as follows:

$$DU = \frac{1}{Q_t} \sum_{i=1}^{\infty} \frac{i * CF_t}{(1 + j_t)^i}$$

Given the payment structure in our model, this will be:

$$DU = \frac{1}{Q_t} \left(\frac{1}{1 + j_t} + \frac{2\delta}{(1 + j_t)^2} + \frac{3\delta^2}{(1 + j_t)^3} + ... \right) = \frac{1 + j_t}{1 + j_t - \delta} \tag{7}$$

As short-term nominal interest rate increases, present-value of future cash flows falls and with it the price of the present bonds. Yield rises because same coupon payments can now be obtained by paying less, and thus duration falls. This can be see in the following log-linearised equation:

$$du_t = -\frac{(1 - \beta)\delta}{1 - \beta\delta} j_t \tag{8}$$

- P_t is a price index that aggregates the prices $(P_j t)$ set by each monopolistic firm and is defined as:

$$P_t = \left[\int_0^1 P_{jt}^{1-\epsilon} dj \right]^{\frac{1}{1-\epsilon}} \qquad \epsilon > 1$$

- B_t is the (nominal) bond holdings at time t (issued at time t), W_t is the nominal wage set in relation to the marginal product of labour, D_t are nominal lump-sum transfers net of taxes. In relation to T_t defined later, $-\frac{D_t}{P_t} = T_t$. Unlike a standard RBC model, the household cannot save in physical capital but only in nominal bonds.

Using the log-linearised first-order conditions from section 5, the goods market equilibrium condition $Y_t = C_t$ output, and that natural (flexible-price) level of output is equal to 0 from section 2.5, we derive the following Dynamic IS equation:

$$y_t = E_t y_{t+1} - \sigma \left(\imath_t - E_t \pi_{t+1} \right) \tag{9}$$

and, log-linearising the first-order conditions for consumption-leisure trade-off from section

5gives the following labour supply equation:

$$w_t - p_t = \frac{1}{\psi} n_t + \frac{1}{\sigma} y_t \tag{10}$$

where, all variables are in log-deviations from the steady state because natural rate of output is equal to the efficient, i.e., steady state level of output (see section 2.5). Also, even though we have a government sector contrary to most standard literature on the fiscal sector, we do not have government's consumption or investment purchases. The role of the government is only through the asset prices of fixed income security - bonds. Finally, since we do not have productivity shocks in this model, natural real rate of return will not fluctuate and thus does not appear in the Dynamic IS equation.

The Dynamic IS equation can be iterated forwards to obtain the following:

$$y_t = -\sigma \sum_{k=0}^{\infty} E_t r_{t+k} \tag{11}$$

where,

$$r_t = i_t - E_t \pi_{t+1} \tag{12}$$

is the ex-ante real interest rate. Thus, output gap depends on path of future ex-ante real interest rates. This will be useful for the description of transmission mechanism in later sections.

2.2 Firms

A standard New Keynesian DSGE introduces two types of rigidities through the firm sector of the economy. Since the goods are differentiated, its producers exercise a monopolistic power and are price setters. This is an important friction in a New-Keynesian economy as it leads to prices being marked up from their marginal costs and wages being marked down, thus leading to output levels being lower than efficient. A second source of friction, which sets such models apart from the typical real business cycle models, is stickiness in prices which is a form of nominal rigidity. Prices follow a Calvo (1983) setting, i.e., there is fixed probability with which a firm can change its price in any given period. It is not

time or state dependent. A monopolist firm can change its price if it is chosen to do so by the Calvo fairy and thus they choose a price taking into account that with probability θ they will not be able to change it again. The optimisation problem of all firms that change their prices in a given time period is the same and they choose a price (P_t^*) that maximises their real profits, discounted by a stochastic discount factor, subject to the demand curve they face from the households and the production technology constraint:

$$E_t \sum_{i=0}^{\infty} \theta^i \Delta_{i,t+i} \left[P_t^* Y_{i,t+i} - \varphi_{t+i}(Y_{i,t+i}) \right]$$

such that

$$Y_{i,t+i} = \left(\frac{P_t^*}{P_{t+i}} \right)^{-\epsilon} C_{t+i}$$

and

$$Y_{i,t+i} = N_{t+i}^{1-\alpha}$$

where $Y_{i,t+i}$ is the output of firms in period $t+i$ if they reset prices in period i and $\varphi_{t+i}(Y_{i,t+i})$ is the nominal total cost for this level of output. Optimal price choosen pins down the output sold by each monopolist firm (through the demand curve) and that pins down the labour demand (through the production function). In the labor market equilibrium, real wage rate is determined based on this labor demand and the labour supply (equation 10). The stochastic discount factor used to find the present value of cash flows of the firm is calculated as

$$\Delta_{i,t+i} = \beta^i \left(\frac{U'_{C,t+i}}{U'_{C,t}} \frac{P_t}{P_{t+i}} \right)$$

Following the algebra as in Galí (2015) (log-linearising the optimal price reset equation, finding an expression that relates firm's marginal cost to the average marginal costs, and find a relation between economy's average marginal cost and output gap, and finally using an expression relating inflation to reset prices) we can find the New-Keynesian Phillip's Curve as follows:

$$\pi_t = \kappa y_t + \beta E_t \pi_{t+1} \tag{13}$$

where, κ, the slope of the Phillip's curve is one of the most important parameters of the New Keynesian DSGE model. It measures the senstivity of inflation to output gap in the economy, which depends on:

(a) Nominal rigidities: Arising from the inability of all firms to change prices in response to changes in the economic conditions, with prices staying same for average $1/(1-\theta)$ months;

(b) Firm-level real rigidities: Price elasticity of the firm's differentiated good's demand (ϵ) arising due to monopolistic competition, and implied elasticity of the marginal cost with respect to the firm's output ($\eta = \alpha/(1-\alpha)$)) arising due to decreasing returns to scale;

(c) Aggregate real rigidities: Elasticity of intertemporal substitution (σ) arising from household's preferences for consumption smoothing over time, i.e., how the household changes its savings decisions in response to changes in ex-ante real interest rates, and Frisch elasticity of labour supply ψ capturing household's trade off between leisure and consumption in response to changes in real wages.

2.3 Government

I use Sheedy (2022)'s discussion on fiscal constraint faced by the government sector:

$$T_t + \frac{Q_t B_t}{P_t} = \frac{(1+\delta Q_t)B_{t-1}}{P_t} \tag{14}$$

where,

- T_t is the real lump-sum tax revenue received by the fiscal authority from the households, net of lump-sum transfers it makes to them. To keep the discussion on monetary policy response more tractable, we abstract away from including the tax revenue dynamics in the model. Instead, we include T_t in the model such that $\phi_t = (\bar{T} - T_t)/\bar{D}$ is a fiscal shock following an i.i.d process, with mean zero[1]. Thus,

[1] $\bar{T}$ and $\bar{D}$ are steady state net taxes and debt levels, See section 2.5

since a shock that changes T_t (either through changes in tax revenues or transfers or both) at time t is one-off, it means that the government cannot raise taxes in the future to finance a positive fiscal shock or pass on a negative shock as higher future transfers. This means that fiscal policy will have no role to play in our model because tax collection cannot be adjusted. Modelling taxes in this way helps us capture the constraints that governments faced after the Great Recession of 2008 and more recently, after COVID, when transfers and debt levels are manifold higher and not possible to be financed through future taxes.

- Q_t is the price of a bond issued in time t and defined as in equation 2. B_t is the number of government bonds outstanding at end of time t. So, essentially, left hand side is total real revenue of the government (taxes and value of bonds carried forward), while right hand side of the constraint captures the first payment made in lieu of outstanding bonds $(1 * B_{t-1})$ plus "buy-back" of outstanding debt at current market value $(\delta Q_t B_{t-1})$[2].

The above constraint can be expressed as follows:

$$T_t + D_t = (1 + R_t)D_{t-1} \tag{15}$$

where, $D_t = \frac{Q_t B_t}{P_t}$ is the real value of government debt at end of time t and R_t is ex-post real return received for holding a bond between periods $t-1$ and t and is therefore defined as:

$$1 + R_t = \frac{(1 + \delta Q_t)/P_t}{Q_{t-1}/P_{t-1}} \tag{16}$$

After log-linearising equation 15 around the respective steady states (see section 2.5), we obtain the following:

$$\beta d_t = d_{t-1} + R_t + \beta \phi_t \tag{17}$$

We iterate it forwards, use the transversality condition $\lim_{t\to\infty} \beta^t d_t = 0$, and that we start

[2]Given the payments due from a bond, one bond issued at time $t-1$, is equivalent to δ new bonds worth Q_t at time t

from the steady state, i.e., $d_{t-1} = 0$ to obtain the following useful equation:

$$-\beta\phi_t = R_t + E_t \sum_{T=0}^{\infty} \beta^{T+1} r_{t+T} \tag{18}$$

Also, log-linearising equation 16 and using equation 3, we have the following relationship between current inflation, and current and past yield rates (see section 5):

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{1-\beta\delta}(j_t - j_{t-1}) \tag{19}$$

2.4 Ramsey Policy

Central Bank follows a policy of commitment, i.e., it minimizes the loss subject to all the constraints (including the fiscal constraint). This is therefore as if fiscal and monetary authorities function as one unit and are not independent of each other. Central Bank chooses a path of $\{y_t, \pi_t, d_t\}_{t=0}^{\infty}$ to minimise a utility-based loss function:

$$L_t = \frac{1}{2} \sum_{l=0}^{\infty} \beta^l E_t \left[\frac{\epsilon}{\nu\xi} \pi_{t+l}^2 + \gamma y_{t+l}^2 \right] \tag{20}$$

subject to following constraints given by equations 4, 9, 12, 13, 17, and 19. The weight put by the Bank on stabilizing output gap γ depends on aggregate-level real rigidity and weight on inflation fluctuations depends on price elasticity (ϵ), nominal rigidities (ξ), and firm-level real rigidities (ν).

2.5 Steady State

Loss function defined as in equation 20 assumes that economy's steady state is efficient and loss is calculated as deviations from it. Steady state values in our model are as follows (notations are consistent with their definitions above, with steady state values being denoted by a bar):

$$\bar{R} = \bar{r} = \bar{j} = \frac{1}{\beta} - 1$$

$$\bar{\pi} = 0$$

$$\bar{y} = 0$$

$$\bar{T} = \bar{R}\bar{D}$$

$$\bar{du} = \frac{1}{1 - \beta\delta}$$

3 Calibration

At the baseline, I calibrate the model as per the United States economy. In macroeconomic literature, parameter values can either be chosen such that steady state matches with the long-run features of the economy, or they can be based on microeconometric studies. They can also be estimated by looking at untargeted moments.

I consider the time unit to be one month. In standard literature, it is generally calibrated to a year or a quarter to match the frequency at which GDP data is publicly released. However, since I want to consider the effect of the duration of government debt, of which Treasury bills are an important component, I calibrate the parameters to a month. In addition, I assume that the economy is in a deterministic/ non-stochastic steady state before the fiscal shock hits the economy.

The baseline calibration is as in table 1. For each parameter, I employ the most commonly used values. For the sensitivity analysis in section 4, I see how optimal policy changes and welfare costs differ as I use the range of values in existing literature.

3.1 Subjective discount factor

This is the rate at which agents discount their utility in every period (here, a month). For the discount rate, I follow Bilbiie et al. (2014) as they calibrate the parameters close to the US economy during the Great Recession of 2007. The authors calibrate the discount rate at 0.9983 (following posterior-mode estimates of Justiniano et al. (2013)) which implies that the annual steady-state ex-ante real interest rate ($\bar{r}$) will be approximately 2% because in steady state: $\beta = (1/1 + \bar{r})^{1/12}$. Note that, in our model, steady state values of ex-post

real return and yield to maturity are equal to steady-state ex-ante real interest rate. 2% average yield is consistent with data as well (averaged for bonds with maturity upto 10 years from Bloomberg).

3.2 Degree of price stickiness (Calvo probability)

For the measure of price stickiness, I follow the source of Schmitt-Grohé and Uribe (2004). The authors use Sbordone (2002)'s estimate that prices tend to change after approximately 9 months. I therefore set $\theta = 8/9$, so that the duration of a price is equal to 9 months because duration $= 1/(1-\theta)$. Although this study uses data before the Great Recession of 2008, and for most of the parameters below I use parameters that corroborate to data in the last two decades, I cross verify that this duration is still consistent (on average) (Nakamura and Steinsson, 2008). This duration is based on the survey of firms about frequency of changes in price conducted by Blinder (1994). Macroeconomic estimates of price stickiness tend to be higher than the estimates based on micro data (Cagliarini et al., 2011). I therefore use micro estimates from Bils and Klenow (2004) who use Bureau of Labour Statistics (BLS) data to find that average duration for prices is 4.3 months for the sensitivity analysis. I also use the highest average macro estimate available of 15 months (Dennis, 2006).

3.3 Elasticity of intertemporal substitution

Engelhardt and Kumar (2009) use evidence from 401(k) participation to arrive at an estimate for the elasticity of intertemporal substitution. I calibrate the parameter to their mean estimate of 0.74 in the baseline model. The authors constructed a 95% confidence interval (0.37–1.21) of their estimate. In section 4, I consider this range for the sensitivity analysis.

3.4 Capital share in production (fixed factor)

To calibrate returns to scale, I follow Bayer et al. (2019), who use the U.S. National Income and Product Accounts (NIPA). They find that the labor's share in income is about 2/3. So, they set the returns to labor at 0.73 after accounting for profits. In light of this, I set

$\alpha = 0.27$.

3.5 Dixit-Stiglitz demand elasticity (for markups)

For the constant elasticity of substitution between the differentiated goods, which corresponds to markup over average marginal costs in the steady state, I use Billi et al. (2021). The authors use latest US data to calibrate the elasticity to equal 9, thus implying the markup in steady state to be 12.5%. In section 4, I will follow Dang (2019) and see if their estimated markups of 63% (with price elasticity estimated at 2.58) change the nature of our analysis.

3.6 Frisch elasticity of labor supply

In Bilbiie et al. (2014) the authors recommend using 0.423 for the Frisch elasticity because it is a mid-way between the relatively low values of the elasticity suggested in microeconometric (labor economics) literature and the high values that are typically used in DSGE (standard business cycle and new keynesian) literature (Chetty et al. (2011) talk about this divergence in detail).

3.7 Duration of government securities

To calibrate duration and thus payment sequence of bonds, I refer to Macaulay duration statistics of 2010 accessed on the OECD database. Here, the duration is stated as 4 years. I use this figure to calibrate δ using the formula shown in table 1. Additionally, for sensitivity analysis I refer to the range of duration estimated by Greenwood et al. (2015). They have shown that average duration has moved very little between 2007 and 2014, from 4 years to 2.9 years. We see how this can alter the costs of optimal policy. So, we consider 2.9 years and the highest duration quoted as 7.8 years. These duration estimates cover what is present in Doepke and Schneider (2006). Since duration is measured in years and our analysis is based on months, we multiply it by 12 to obtain a monthly figure, before estimating δ.

4 Sensitivity Analysis

4.1 Optimal Policy as duration of debt changes

Substituting equations 19, either 5 or 6, and 12 in equation 18, I arrive at equation 21 where it is easy to see that a one-off fiscal shock can be financed by a combination of - inflation in current period, or future expected inflation, or changes in current and future short-term nominal interest rates which then affect the time path of bond prices and yield to maturity. In a Ramsey policy, where the path of future inflation is decided and the Central Bank commits to it, inflation expectations are well-anchored. Equation 18:

$$-\beta\phi_t = R_t + E_t \sum_{T=0}^{\infty} \beta^{T+1} r_{t+T}$$

$$-\beta\phi_t = -\pi_t + E_t \sum_{T=0}^{\infty} \beta^{T+1}(1-\delta^{T+1}) i_{t+T} - E_t \sum_{T=0}^{\infty} \beta^{T+1} \pi_{t+1+T} \tag{21}$$

When $\delta = 0$, i.e., standard case of one-period nominal bonds, output gap is used to absorb the shock. Since the shock is not persistent, a rational agent expects the economy to be back in the steady state in the next period and hence, expected inflation is zero. If the bank uses sharper movements in current inflation, it will produce very high fluctuation in inflation. To maximise welfare, the Central Bank will reduce the short-term nominal interest rate (which coincides with both ex-ante real interest rate and long-term nominal interest rate when $\delta = 0$), thus inflating bond prices because future cash flows are discounted at a smaller rate and reducing real debt burden of the fiscal authority because less bonds need to be issued for the same level of positive fiscal shock. I will call this channel of using ex-ante short-term interest rates to change bond prices - term structure channel. (See figure 1).

In contrast, when $\delta = 1$, i.e., a perpetuity, output gap hardly moves. Inflation jump in the period of the shock and a fall in ex-post real return due to those who held bonds in the period before shock helps finance a portion of the shock, while a constantly high (although decaying) inflation thereafter helps reduce the real debt burden of the government. Thus

existing and future bond-holders help absorb the shock forever. I will call this channel of using current and future (expected) inflation to change ex-post interest rates - inflation channel.

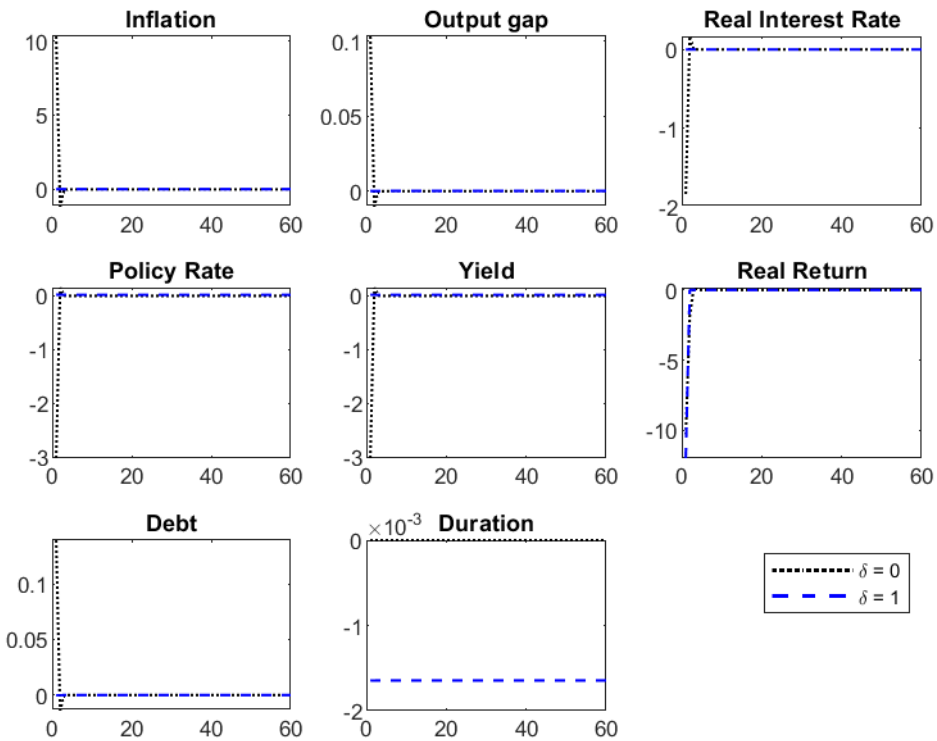

Figure 1: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for δ. This helps us to see the contrast between welfare from inflation and output gap smoothing motives, for a given level of price stickiness - 9 months. Note that, it may not be clear from the figure but in $\delta = 1$ case, inflation jumps to a high steady state of 0.02.

These extreme cases gives us a sense of the trade-offs the Ramsey planner faces as the duration of debt changes. Firstly, duration helps the Central Bank/ Ramsey planner to smooth out fluctuations in output gap and inflation to absorb the shock. Since the welfare maximising function depends on fluctuations in the two variables and not levels, a higher duration helps the planner to smooth out inflation. Even though the variables take more time to converge, especially inflation, it is not welfare reducing because fluctuations far in the future are discounted in the loss function. Hence, fluctuations away from the steady state closer in time are more welfare reducing than such fluctuations far into the future. Hence, as duration rises, the initial jump is less for both the inflation and output

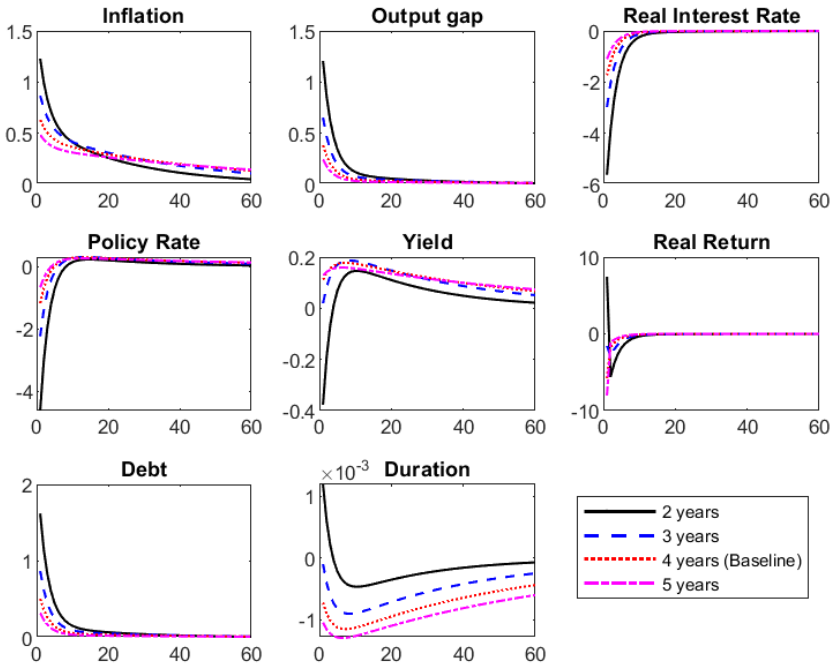

Figure 2: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for δ. This helps us to see the effect of higher duration on welfare from inflation and output gap smoothing, for a given level of price stickiness - 9 months.

gap. (See figure 2)

As duration increases, the role of the term structure channel weakens and we see smaller declines in the policy rate (i.e., short-term nominal rates) on impact (i.e., at time of the shock) as well as after that until it converges to the steady state. Thus, bond prices rise less and absorb less of the debt burden as duration increases due to equation 6. This is welfare improving strategy because a path of higher bond prices will not erode the real market value of debt by as much when δ is higher. Hence, the role of inflation channel intensifies with higher duration and debt burden is reduced gradually through inflation above the steady state for longer time than when duration is lesser, i.e., higher duration helps to smooth inflation over time.

4.2 Optimal Policy when nominal rigidity interacts with duration of debt

When we have higher price rigidity (i.e., higher θ), weight assigned to inflation stability in the loss function increases (because measure of nominal rigidities, ξ decreases) and the relative weight assigned to output stabilization decreases. Thus, the magnitude of increase in surprise inflation after a positive fiscal shock is lesser when we have sticky prices. For a lower duration, if prices are flexible, using the surprise inflation is more optimal because output gap is always 0 (figures 2 and 3). But as price stickiness increases, given the same duration, welfare costs associated with using the inflation channel increase and inflation fluctuations can be traded-off for output gap fluctuations (figure 4). Again, for a given level of price stickiness, higher duration reduces loss value, i.e., improves welfare when inflation channel is used such that surprise inflation is low but the path is smoothed out through higher expected inflation (see table 2). Interest rates rise (more if shorter duration) and are above the steady state when flexible prices because output gap needs to be zero, which is in contrast to fall in the rates in case of sticky prices. Because of sharp surprise inflation, overall debt levels are low in flexible prices. Figure 4 shows the trade offs as rigidity increases for a given duration. When prices are flexible, output gap is more important and we therefore see sharper movements in inflation. However, higher duration in this case (4 years), helps to improve welfare by smoothing out inflation when prices are sticky and thus the use of term structure channel is welfare improving.

4.3 Optimal Policy with different returns to scale (firm-level real rigidity)

η measures the implied elasticity of the marginal cost with respect to the firm's output ($\eta = \alpha/(1-\alpha)$)) and is positively related with α because higher the α, more the returns are decreasing in labour, and hence more labor hours need to be employed to produce the same marginal output, and hence higher the increase in marginal cost. Thus, for a given output gap, inflation should respond more strongly but prices are sticky.

When we have higher η, weights assigned to both inflation and output gap stability

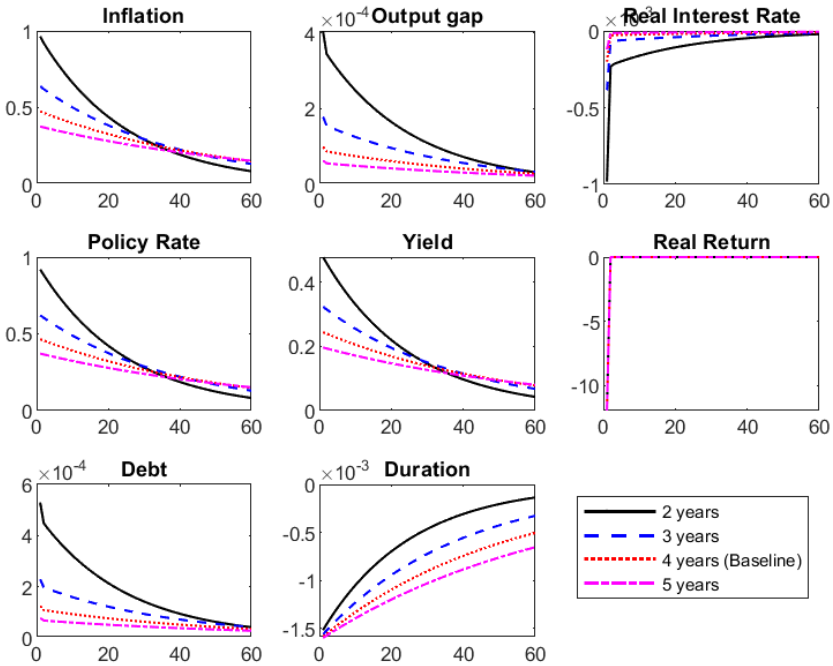

Figure 3: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for δ. This helps us to see the effect of higher duration on welfare from inflation and output gap smoothing, for a given level of price stickiness - 1 month (close to having flexible prices).

in the loss function increase. Which of these is more important depends on calibration of other parameters. For my baseline calibration, it seems that stabilising inflation is more important. Since prices are sticky, the Central Bank can use the term structure/ output gap channel to finance the fiscal shock and reduce the burden of debt, without paying the cost of excess inflation. Hence, it reduces policy rate and thus the ex-ante real rate as expected inflation does not increase by much, which pushes up the bond prices and helps finance the debt without using very high inflation. Short-term nominal rates can fall more for higher η because it implies higher firm-level rigidity and thus lower optimal price/ inflation response. (See figure 5.) Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when returns are decreasing at a faster rate (table 3).

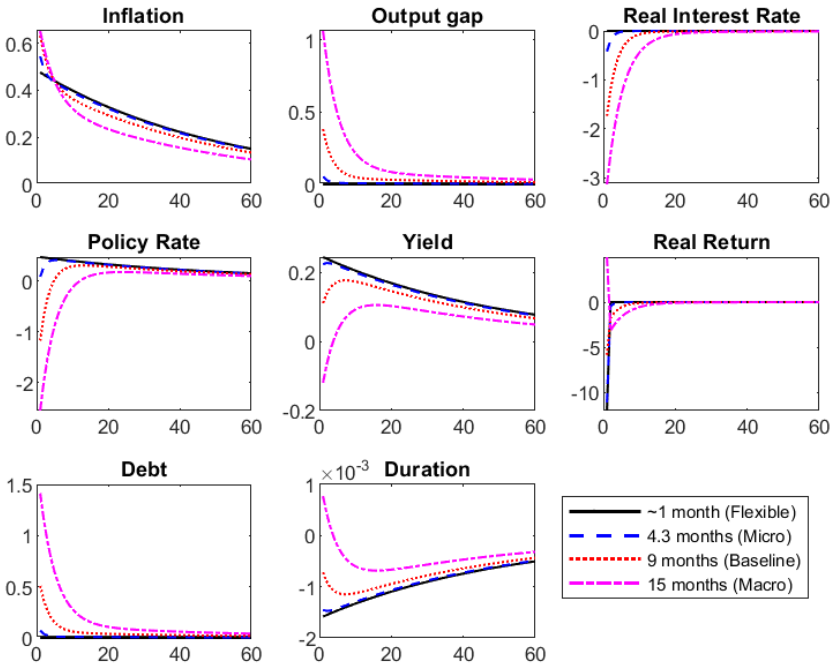

Figure 4: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for θ. This helps us to see the effect of higher price stickiness on welfare from inflation and output gap smoothing, for a given duration - 4 years.

4.4 Optimal Policy with different markups (firm-level real rigidity)

When price elasticity of the firm's differentiated good's demand (ϵ) arising due to monopolistic competition is low, i.e., when markups μ are higher, weights assigned to inflation stability in the loss function decreases because for a given output gap, higher mark-ups (lower ϵ) would mean sharper movements in inflation and thus to lower fluctuations in both output gap and inflation, it is welfare maximising to target output gap volatility. Consequently, Central Bank uses surprise inflation (lower ex-post real returns) and higher but smoothed out inflation expectations to finance the fiscal shock. We are able to do the latter because of higher duration. Welfare losses when duration is low but markups are high, are high because this additional channel of smoothing inflation is not available (table 4) . Therefore, in contrast to the returns to scale case, policy rate and ex-ante real rate and thus the output gap fluctuate less, when higher the markup (figure 6).

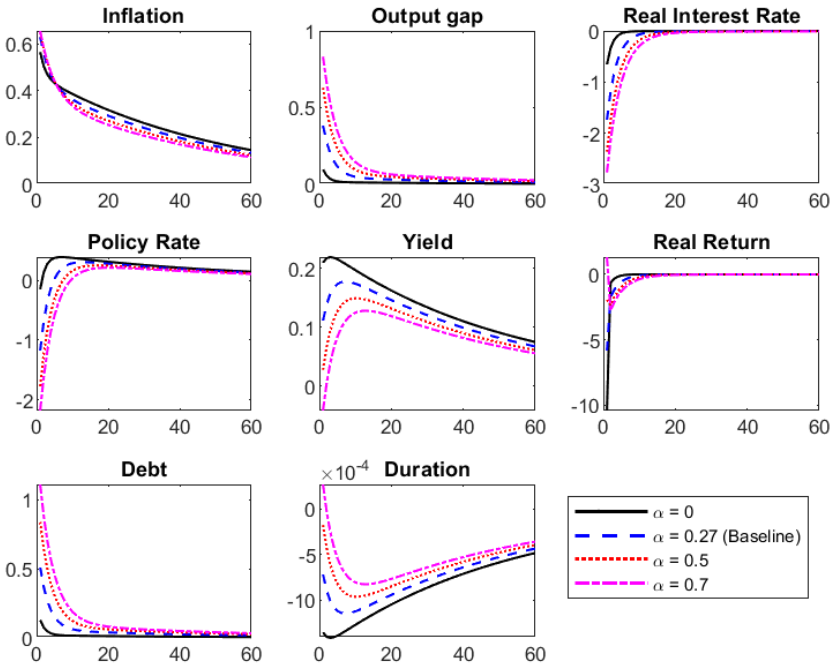

Figure 5: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for α. This helps us to see the effect on welfare from inflation and output gap smoothing, for a given duration - 4 years and price stickiness - 9 months.

4.5 Optimal Policy with different EIS (aggregate real rigidities)

Higher the elasticity of intertemporal substitution (σ), higher is the risk appetite of households. The term structure channel becomes less useful to control fluctuations when EIS is higher, because changes in the ex-ante real interest rates (due to a combination of changes in short-term nominal interest rates and smoothening out of expected inflation), implies sharper movements in output gap and thus inflation because of higher propensity to move consumption over time. Thus, we see an increased use of surprise inflation and higher future inflation to finance the fiscal shocks, instead of the bond prices channel for higher EIS. Note that yields here are above the steady state despite policy rates being below the steady state initially, because yield in time t is a function of the entire future path of nominal interest rates, not just the contemporaneous rate (figure 7). Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when EIS is high (table 5).

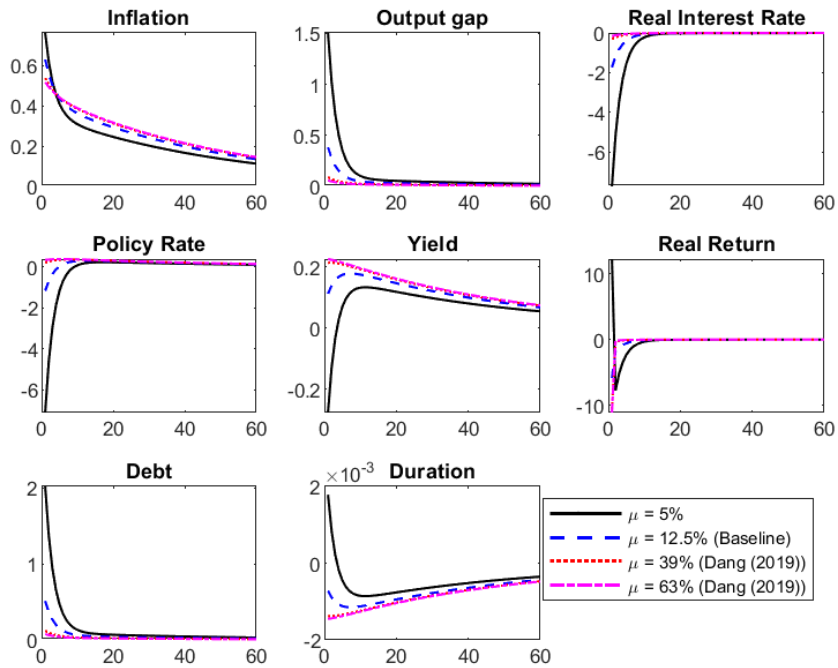

Figure 6: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for ϵ. This helps us to see the effect on welfare from inflation and output gap smoothing, for a given duration - 4 years and price stickiness - 9 months.

4.6 Optimal Policy with different Frisch elasticities (aggregate real rigidities)

Frisch elasticity of labour supply ψ measures percentage change in labour supply by the household in response to 1% change in real wages. If the supply increases in response to increase in wages, we say that the substitution effect (higher wage implies higher opportunity cost of leisure) dominates the income effect (higher wage implies higher income for same labor hours worked). Frisch elasticity plays a role in how much economy's average marginal cost should increase (higher wages) to incentivise households to work more. If the elasticity is higher, response of costs and hence inflation to a given output gap is smaller. As people are more willing to work when ψ is higher, a positive output gap would enable them to earn more income which can then be used to finance the fiscal deficit through higher bond prices. Therefore, fall and overall path of policy rates and hence, yields is lower for higher ψ (since bond price varies inversely with the whole path of future short-term nominal interest rates) (figure 8). Higher duration reduces the loss

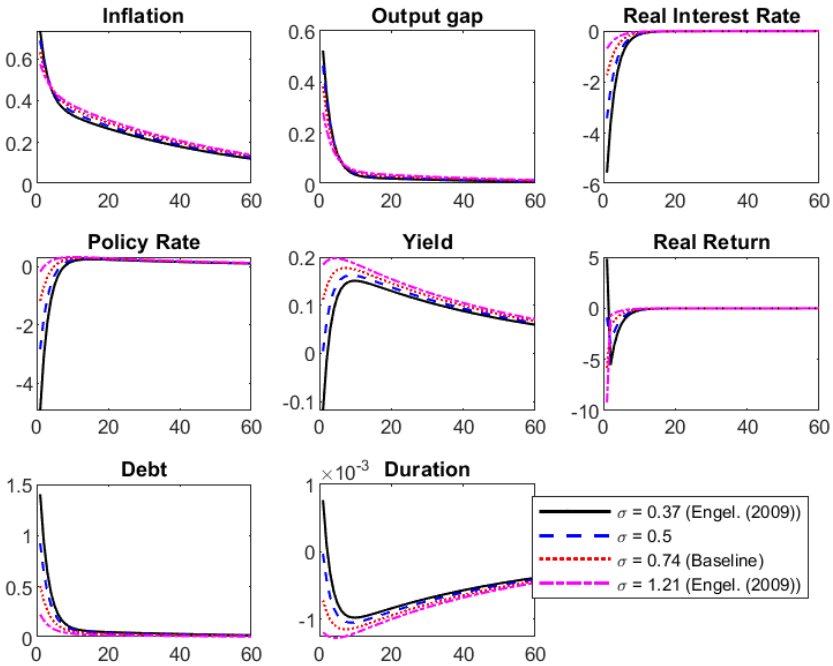

Figure 7: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for σ. This helps us to see the effect on welfare from inflation and output gap smoothing, for a given duration - 4 years and price stickiness - 9 months.

value because it helps to smooth out inflation over time. This effect is stronger and more helpful when elasticity is low (table 6).

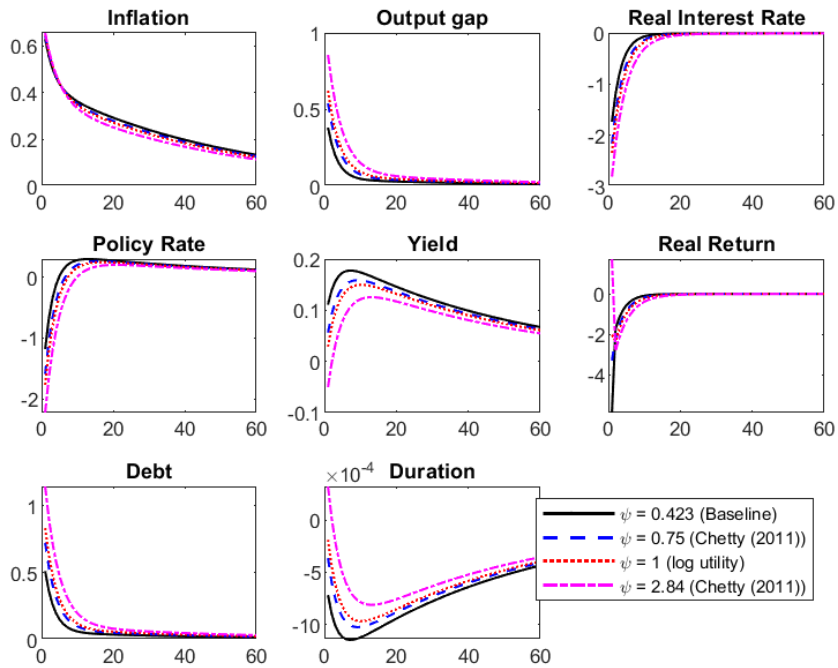

Figure 8: IRFs for 1 standard deviation fiscal shock for a model calibrated as per 1, except for ψ. This helps us to see the effect on welfare from inflation and output gap smoothing, for a given duration - 4 years and price stickiness - 9 months.

5 Appendix

5.1 Household's Optimisation

First order conditions:

With respect to C_t and L_t are same as in basic case, With respect to B_t is as follows:

$$\beta^t(\lambda_t Q_t) = \beta^{t+1} E_t \left(\lambda_{t+1}(1 + \delta Q_{t+1})\right)$$

Euler equation is thus:

$$Q_t = \beta E_t \left[\left(\frac{C_t}{C_{t+1}} \right)^{1/\sigma} \frac{P_t}{P_{t+1}} (1 + \delta Q_{t+1}) \right]$$

Log-linearise this around the steady state:

$$\sigma q_t = c_t - E_t c_{t+1} - \sigma E_t \pi_{t+1} + \sigma \beta \delta E_t q_{t+1}$$

where $x_t = logX_t - logX$ and $Q = \frac{\beta}{1-\beta\delta}$

Log-linearising equation 2 around the steady state gives equation 3. Using equation 3 here gives:

$$c_t = E_t c_{t+1} - \sigma \left(-E_t \pi_{t+1} - q_t + \beta \delta E_t q_{t+1}\right)$$

i.e.,

$$c_t = E_t c_{t+1} - \sigma \left(-E_t \pi_{t+1} + \frac{1}{1-\beta\delta}(j_t - \beta\delta E_t j_{t+1})\right)$$

First order condition with respect to B_t^k gives the standard Euler equation as follows:

$$c_t = E_t c_{t+1} - \sigma \left(-E_t \pi_{t+1} + i_t - \rho\right)$$

Upto an approximation, equating the two Euler equations under no-arbitrage conditions gives us what we can obtain from the Expected theory of interest rates as well:

$$\left(-E_t \pi_{t+1} + \frac{1}{1-\beta\delta}(j_t - \beta\delta E_t j_{t+1})\right) = \left(-E_t \pi_{t+1} + i_t\right)$$

$$i_t = \frac{1}{1-\beta\delta}(j_t - \beta\delta E_t j_{t+1})$$

which gives equation 4.

5.2 Ex-post real return

Log-linearising equation 16 around a deterministic steady state gives:

$$r_t = -\frac{\pi_t}{1-\beta} - \frac{q_t}{1-\beta} + \frac{\delta\beta}{1-\beta} q_{t+1}$$

Substituting equation 3, we get equation 19:

$$r_t = j_{t-1} - \frac{\pi_t}{1-\beta} - \frac{\delta\beta}{1-\delta\beta}(j_t - j_{t-1})$$

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Table 1: Baseline Calibration

Description	Symbol	Definition	Value
Subjective discount factor	β		0.9983
Degree of price stickiness (Calvo)	θ		8/9
Elasticity of intertemporal substitution	σ		0.74
Capital share in production (fixed factor)	α		0.27
Elasticity of MC wrt firm output	η	$\alpha/(1-\alpha)$	0.369
Dixit-Stiglitz demand elasticity	ϵ		9
Frisch elasticity of labor supply	ψ		0.423
Payment sequence of government bonds	δ	$(1/\beta)(1-(1/\chi))$	0.9808
Macaulay duration of government bonds	χ		4*12
Measure of firm-level real rigidities	ν	$1/(1+\epsilon\eta)$	0.231
Measure of aggregate-level real rigidities	γ	$(1/\sigma)+((1+\eta)/\psi)+\eta$	4.959
Measure of nominal rigidities	ξ	$(1-\theta)(1-\beta\theta)/\theta$	0.014
Gradient of Phillips Curve	κ	$\nu\gamma\xi$	0.016

Macaulay Duration	**Loss value:** $\theta = \mathbf{0.1}$	**Loss value:** $\theta = \mathbf{0.89}$
One-period	0.0182	0.0148
2 years	0.0019	0.7516
3 years	0.0013	0.6268
4 years (Baseline)	0.0095	0.5046
5 years	0.0075	0.414
7.8 years (Greenwood et al (2015))	0.0047	0.2667
Perpetuity	0.00002	0.0128

Table 2: Loss values are in % of the steady state consumption. Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when prices are more sticky.

α		**Loss value:**	
	dur = 1 years	**dur = 4 years**	**dur = 7.8 years**
0.1	0.4433	0.1237	0.1241
0.27 (Baseline)	0.5946	0.5046	0.2667
0.5	0.8506	1.005	0.6095
0.7	1.365	2.2945	1.327

Table 3: Loss values are in % of the steady state consumption. Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when returns are decreasing at a faster rate.

μ		**Loss value:**	
	dur = 1 years	**dur = 4 years**	**dur = 7.8 years**
5%	0.505	2.0748	1.3909
12.5% (Baseline)	0.5946	0.5046	0.2667
39% (Dang (2019))	0.225	0.1092	0.0332
63% (Dang (2019))	0.1182	0.0659	0.0158

Table 4: Loss values are in % of the steady state consumption. Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when markups are high.

σ		**Loss value:**	
	dur = 1 years	**dur = 4 years**	**dur = 7.8 years**
0.37 (Engelhardt et al (2009))	0.2861	0.467	0.2632
0.5	0.3951	0.4854	0.2649
0.74 (Baseline)	0.5946	0.5046	0.2667
1.21 (Engelhardt et al (2009))	0.9344	0.5216	0.2683

Table 5: Loss values are in % of the steady state consumption. Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when EIS is high (1.21).

ψ		**Loss value:**	
	dur = 1 years	**dur = 4 years**	**dur = 7.8 years**
0.423 (Baseline)	0.5946	0.5046	0.2667
0.75 (Chetty (2011))	0.4286	0.4864	0.2649
1 (log utility)	0.3716	0.477	0.2639
2.84 (Chetty (2011))	0.2576	0.4484	0.2608

Table 6: Loss values are in % of the steady state consumption. Higher duration reduces the loss value because it helps to smooth out inflation over time. This effect is stronger and more helpful when elasticity is low.